

Exam - Object-Oriented Programming

Emily Antosch
HAW Hamburg · Faculty TI

Winter semester 2026/27 · 90 minutes · 105 points

Theory: 60 minutes (Tasks 1–5, 65 points). Programming: 30 minutes (Task 6, 40 points).

Name:	Student ID:
Date:	

Instructions

- Work individually. Answer Tasks 1–5 in the spaces provided; label any additional sheets with your student ID and task number.
- For Task 6, submit the Java source files in a folder named StudentID_LastName. Use a standard JDK (Java 17 or later), without preview features or external libraries.
- No GUI, interactive menu, file storage, or concurrency is required. Each task is independent.
- You may use two double-sided A4 sheets or four single-sided A4 sheets as your cheat sheet. Only handwritten notes, made with a pen or on a tablet, are permitted. Typed and printed cheat sheets are not allowed.
- Submit both your paper and your programming tasks using the designated drive in Explorer.
- You may start with either theory or programming and switch between them at any time.
- Check that your paper is complete using the page count in the footer.

Tasks

Task 1: Basic Concepts	2
Task 2: Object-Oriented Reasoning	3
Task 3: True or False	4
Task 4: Code Completion	5
Task 5: Code Debugging	6
Task 6: Room Sensor Log	7

Task	1	2	3	4	5	6	Total
Available	10	15	16	12	12	40	105
Achieved							

THEORY · 60 MINUTES IN TOTAL

Task 1: Basic Concepts

10 points

For each situation, explain the relevant Java concept and give one short Java example. Each item earns **1 point for the explanation and 1 point for the example**.

a) Object references (2 points). Two variables should refer to the same Lamp object rather than two separate lamps. Explain what reference assignment copies.

b) Encapsulation (2 points). A thermostat must reject target temperatures outside its allowed range. Explain why its temperature field should not be public.

c) Constructors (2 points). A Room must receive its name when it is created. Explain the constructor's role and show the constructor declaration.

d) Static members (2 points). All Room objects share the same maximum permitted occupancy. Explain why a class constant is suitable.

e) Immutable strings (2 points). Explain why calling `name.toUpperCase()` alone does not change the string stored in `name`. Show how to retain the result.

THEORY · 60 MINUTES IN TOTAL

Task 2: Object-Oriented Reasoning

15 points

a) Inheritance and super (5 points). Given the superclass below, write a subclass `Bicycle` with a constructor `Bicycle(String id)` that initializes the inherited state. Override `label()` to return `Bike`: followed by the superclass `label`. Keep the field private and use `super` in both places.

```
1  class Vehicle {
2      private String id;
3      public Vehicle(String id) { this.id = id; }
4      public String label() { return id; }
5  }
```

Java

b) Overloading and overriding (5 points). Read this separate program. State its two output lines (2 points). Identify the overloaded methods (1 point), the overridden method (1 point), and explain why the first line uses `Bell.sound()` (1 point).

```
1  class Signal {
2      public String sound() { return "beep"; }
3      public String sound(int count) { return "count=" + count; }
4  }
5  class Bell extends Signal {
6      @Override
7      public String sound() { return "ding"; }
8  }
9  // Inside main:
10 Signal signal = new Bell();
11 System.out.println(signal.sound());
12 System.out.println(signal.sound(2));
```

Java

c) Interface or abstract class? (5 points). Unrelated classes must all offer `double energyUse()`. Give a suitable interface declaration (2 points). A family of appliances additionally shares a private name field, a constructor, and a concrete getter. Explain why an abstract superclass can hold this shared implementation (2 points). Can that abstract superclass be instantiated directly (1 point)?

THEORY · 60 MINUTES IN TOTAL

Task 3: True or False

16 points

Mark exactly one column for each statement. Each correct answer earns **2 points**. A missing, or double-marked answer earns 0 points. An incorrect answer yields **-2 points**; the minimum number of points for this task remains 0.

No.	Statement	True	False
1	A local variable of a class type stores a reference to an object, or <code>null</code> .		
2	Declaring an array reference <code>final</code> prevents changes to its elements.		
3	A subclass inherits its superclass's constructors.		
4	A class can implement two interfaces while extending one class.		
5	A static method can read an instance field if it has a suitable object reference and access is permitted.		
6	<code>String.equals</code> compares the text content of two strings.		
7	The first valid index of a nonempty Java array is 1.		
8	A method declared <code>void</code> must end with <code>return null;</code> .		

THEORY · 60 MINUTES IN TOTAL

Task 4: Code Completion

12 points

Complete gaps **A–F** directly in the listing (**2 points per gap**). WaterTank stores an integer water level from **0 through 100 inclusive**. Invalid constructor or setter arguments must throw `IllegalArgumentException` without changing an existing level. The constructor must reuse the setter. `freeCapacity()` returns `100 - level`. The string format is exactly `WaterTank[level=25]` when the level is 25.

```
1  public class WaterTank {
2      // A: declare the private int field
3
4
5      public WaterTank(int level) {
6          // B: initialize through the setter
7
8      }
9      public int getLevel() {
10         // C
11
12     }
13     public void setLevel(int level) {
14         // D: validate first, then update the field
15
16
17
18
19     }
20     public int freeCapacity() {
21         // E
22
23     }
24     @Override
25     public String toString() {
26         // F
27
28     }
29 }
```

Java

THEORY · 60 MINUTES IN TOTAL

Task 5: Code Debugging

12 points

Find **six independent defects**. For each, identify the location, explain the problem, and give a correction (**1 point for the explanation, 1 for the correction**). The specification requires all instance fields to be private; the constructor to retain its arguments; `hasName` to compare text content; and `sum` to add every array element exactly once. Assume names and arrays are non-null. An empty array must sum to 0. Missing getters, formatting choices, and Java's default field initialization are **not** defects.

```
1  public class ScoreCard {
2      public String name;
3      private int bonus;
4
5      public ScoreCard(String name, int bonus) {
6          this.name = name
7          bonus = bonus;
8      }
9      public static int getBonus() {
10         return bonus;
11     }
12     public boolean hasName(String other) {
13         return name == other;
14     }
15     public int sum(int[] scores) {
16         int total = 0;
17         for (int i = 0; i ≤ scores.length; i++) {
18             total += scores[i];
19         }
20         return total;
21     }
22 }
```

Java

PROGRAMMING · 30 MINUTES

Task 6: Room Sensor Log

40 points

Build a small console application that stores room measurements and calculates a summary. Use three files: `Reading.java`, `SensorLog.java`, and `Demo.java`. You do **not** need to implement keyboard input, editing, searching, or deletion.

Data and behavior

- A `Reading` has a room name (`String`) and an integer temperature in °C (`int`). Once created, it has no methods that change these values.
- The `Reading` constructor receives valid values; validation is the responsibility of `SensorLog.addReading`.
- A valid room name is non-null and not empty after `trim()`. Store the trimmed name. Temperatures must be between **-20 and 50 inclusive**.
- Repeated room names are allowed: they represent separate measurements.
- `addReading(String room, int temperature)` returns `true` after adding a valid reading. Otherwise it returns `false` and leaves the list unchanged.
- `countAbove(int limit)` returns the number of readings **strictly greater than** the limit.
- `averageTemperature()` returns a `double` average over all stored readings. It returns `0.0` when there are none. Avoid integer division.
- `displayAll()` prints each reading in insertion order, or `No readings` when empty.

Implementation and points

Part	Deliverable	Points
a	Create the named project folder and the three Java files with matching public classes.	3
b	<code>Reading</code> : two private fields, constructor, both getters, and a readable <code>toString()</code> containing both values.	5
c	<code>SensorLog</code> : private <code>ArrayList<Reading></code> initialized as an empty list.	4
d	Implement <code>boolean addReading(String room, int temperature)</code> with the specified validation.	6
e	Implement <code>int countAbove(int limit)</code> ; an empty log returns 0.	5
f	Implement <code>double averageTemperature()</code> including its empty-log behavior.	6
g	Implement <code>void displayAll()</code> including the empty message.	3
h	<code>Demo.main</code> : demonstrate the acceptance checks on the next page with labelled output or assertions.	5
i	Use clear names, consistent indentation, and encapsulation; keep log operations in <code>SensorLog</code> , not <code>Demo</code> .	3

Task 6: Acceptance checks and API reminder

Use a fresh log for this sequence:

1. Display it while empty; check that its average is 0.0 and countAbove(20) is 0.
2. Add (" Kitchen ", 20) and ("Kitchen", 21); both calls must return true. The displayed room names must have no surrounding spaces.
3. Check that the average is 20.5, countAbove(20) is 1, and countAbove(21) is 0.
4. Try a null room name, a spaces-only room name, and a temperature of 51; all must return false. The average and counts must remain unchanged.
5. Display the two valid readings in their insertion order.

ArrayList needs import java.util.ArrayList;. These operations may help:

```
1  ArrayList<Reading> readings = new ArrayList<>();
2  readings.add(new Reading("Hall", 18));
3  int size = readings.size();
4  boolean empty = readings.isEmpty();
5  for (Reading reading : readings) {
6      System.out.println(reading);
7  }
8  // For non-null text:
9  boolean blank = text.trim().isEmpty();
```

Submit all three .java files. A compiling partial solution can earn credit for the parts it implements correctly.